NN&DL

Neural Networks Act I: 1943

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Sem. II

Summary

1 Neural Networks: definition

2 The McCulloch and Pitts Neuron (1943)

3 A Logical Calculus

1. Neural Networks: definition

Neural Networks

Neural networks (artificial neural networks, ANN) are computational models inspired by the functioning of the biological brain. They are, without a doubt, the most important and successful technology in AI over the last thirty years.

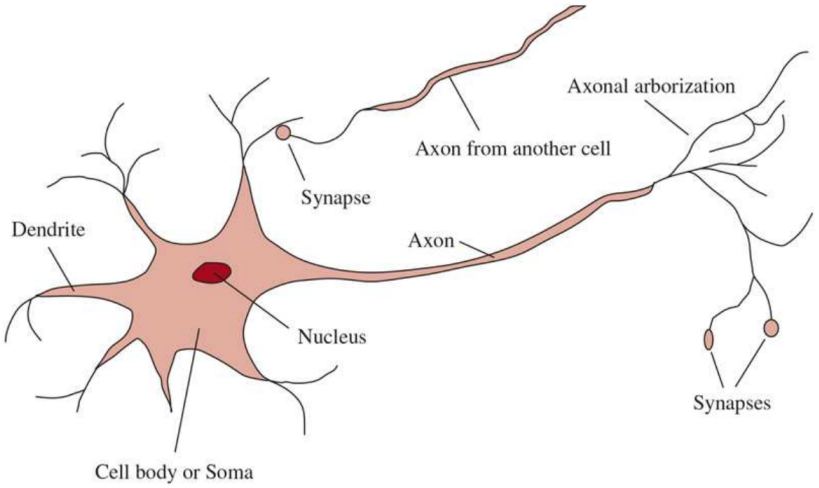
They originate in 1943, even before AI itself was formally established as a scientific discipline (1956), as a (highly simplified) model of the functioning of **biological neurons**.

Neurons

Neurons are the fundamental cells of the nervous system. They were first observed individually by Camillo Golgi (1843–1926) thanks to staining techniques for microscopic analysis.

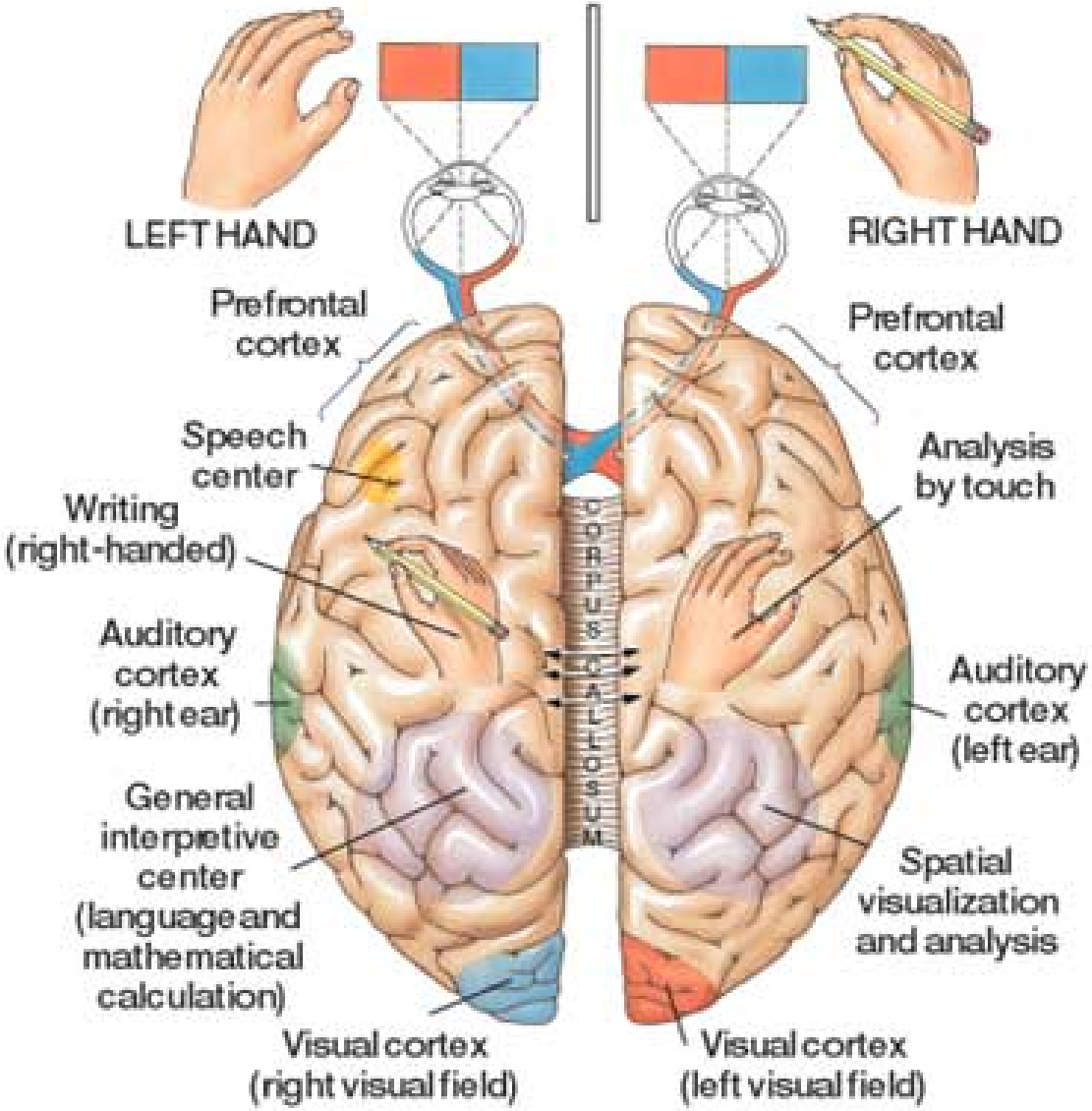
Studies on neuronal organization begin with Santiago Ramón y Cajal (1852–1934). Today we know that cognitive functions are the result of electro-chemical interactions among neurons.

Quoting John Searle (1992): "*brains cause minds*"—we have known since the Second World War that damage to a specific region of the brain leads to the loss of specific cognitive or motor functions.



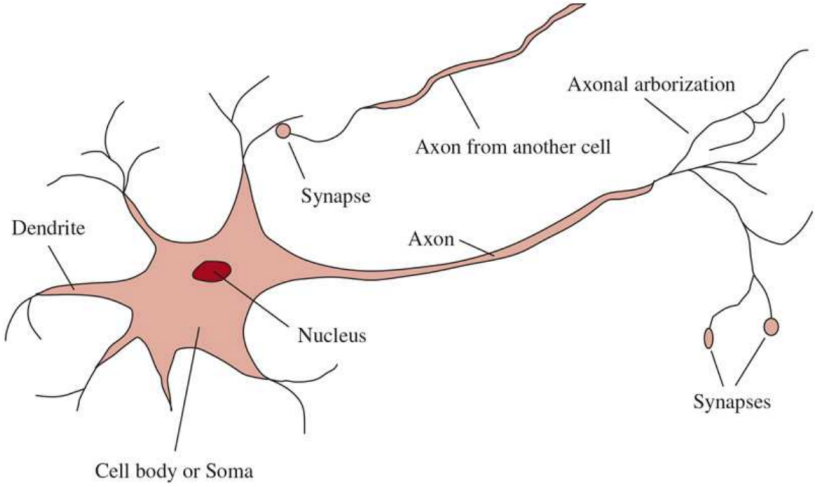
Some regions of the brain and their functions

- Prefrontal cortex: decision control, planning, emotion regulation, attention, social behavior.
- Parietal lobe: integration of sensory information (touch, temperature, pain), spatial perception and orientation, reading and arithmetic abilities.
- Occipital lobe: processing of visual information.
- Temporal lobe: auditory processing, language comprehension, memory (with the hippocampus), object and face recognition.
- Cerebellum: motor coordination, balance, movement precision, motor learning—structurally very similar to an artificial **feed-forward neural network** (FFNN).
- Amygdala: processing of emotions, especially fear, stress, and defensive responses.
- Hippocampus: long-term memory, spatial orientation, learning.
- Thalamus: relay station for sensory information (except olfaction); regulates sleep and wakefulness.
- Hypothalamus: regulation of homeostasis (internal balance), control of autonomic functions (body temperature, hunger, thirst, hormone release).
- Basal ganglia: control of voluntary movements, motor learning, execution of repetitive actions and habits.



https://engineuring.wordpress.com/2010/04/25/consciousness-in-mixed-systems-merging-artificial-and-biological-minds-via-brain-machi

Neurons and their signals



A neuron consists of:

- a cell body (soma) containing the nucleus
- branches (dendrites) that receive signals from other neurons
- a long extension (axon) along which signals propagate
- axon terminals that release neurotransmitters into synapses (the connections between neurons)

How neurons communicate

Signals propagate along the axon as **electrical action potentials**, i.e., brief, all-or-nothing changes in membrane voltage generated by local ion flows across the membrane (ions do not travel along the neuron; the voltage difference does).

When an action potential reaches the axon terminals, it triggers the release of **neurotransmitters** into the synapse, which modulate the membrane potential of nearby neurons.

Neuronal communication involves two distinct physical mechanisms:

- **Within a neuron**, signaling is **electrical**. Action potentials propagate along the axon through successive, local changes in membrane potential caused by ion exchanges.
- **Between neurons**, signaling is **chemical** (in most cases). The arriving action potential causes the release of neurotransmitters into the synaptic cleft.

Neurotransmitters then bind to receptors on the postsynaptic membrane and open ion channels, producing local **depolarizing (excitatory)** or **hyperpolarizing (inhibitory)** effects. The neuron integrates these effects, and a new action potential is generated only if a threshold is reached.

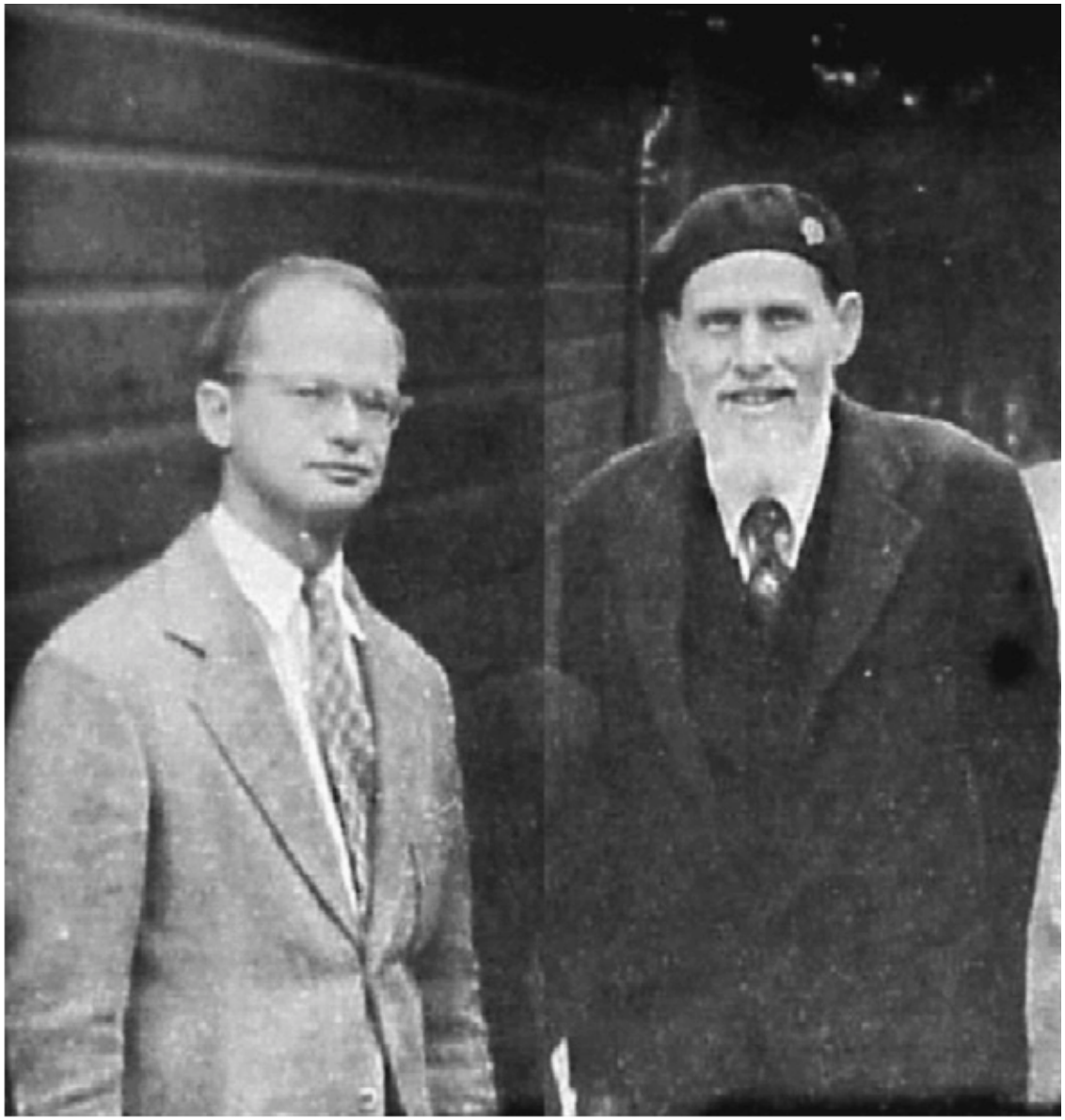
2. The McCulloch and Pitts Neuron (1943)

McCulloch and Pitts

In 1943, Warren McCulloch (a neurophysiologist) and Walter Pitts (a mathematician) proposed a very simple model of the biological neuron based on binary logic.

To do so, they drew inspiration both from studies on the physiology and functions of biological neurons in the brain, and from the work on propositional logic by Russell and Whitehead (1948) and on computation by Alan Turing.

The **McCulloch-Pitts neuron** has two states: on and off. It turns on (activates) when it receives a sufficiently large stimulus from neighboring neurons. Otherwise, it remains quiescent.



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A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY*

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Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

The Artificial Neuron Model

The artificial neuron of McCulloch and Pitts is, in essence, a **computational unit** that receives **numerical signals** $x = (x_1, x_2, \ldots, x_n)$as input, processes them, and produces a numerical signal y as output.

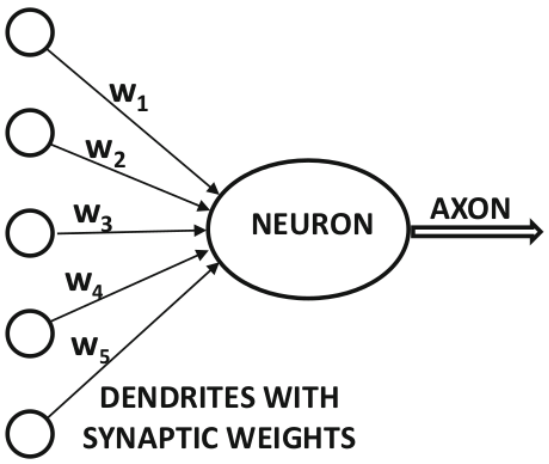
The key element of this computational unit is the **synaptic weights**.

In their model (which we still use today), each pair of connected neurons i, j has an associated **synaptic weight** (denoted w_{ij}) that modulates the influence of the source (upstream) neuron i on the target (downstream) neuron j.

If $w_{ij} > 0$, neuron i has an excitatory effect on j (i.e., it makes j more likely to activate). Conversely, if $w_{ij} < 0$, neuron i has an inhibitory effect on j (i.e., it makes j less likely to activate).

The output y of the neuron is obtained by comparing the total **weighted** signal $p = \sum_{i=1}^{n} w_{ij} x_i$ received by neuron j with an activation threshold (also called the **firing threshold** or **bias**) b. If the total signal p exceeds the threshold, the neuron activates (i.e., produces an output of 1); otherwise, it remains quiescent (output 0).

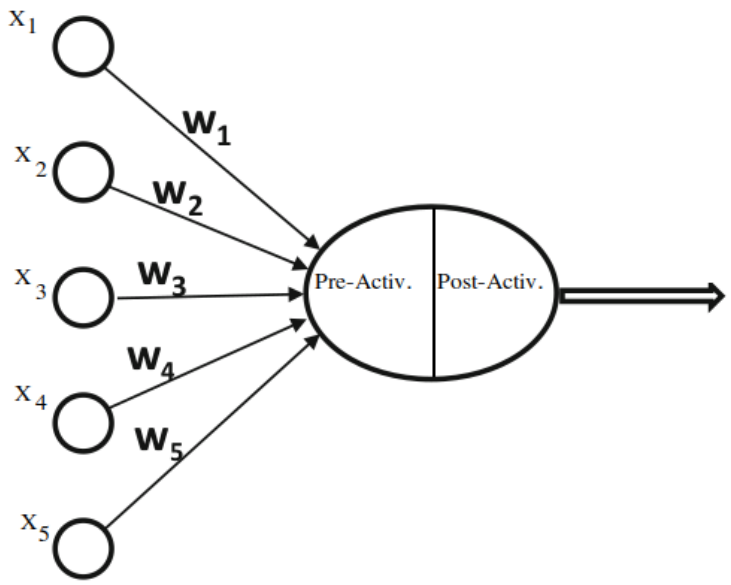
Artificial neurons



In most cases (tranformers, which are key to Large Language Models, contain neurons with a different architecture), the artificial neuron carries out a two-step computation:

- First, it calculates the neuron's **pre-activation**, i.e., the total signal it received from its inputs, weighted by its synaptic weights.
- Second, the pre-activation is passed through an **activation function** Φ, whose role is to determine whether the neuron will be quiescent or excited; in the latter case, Φ also computes the numerical value of the signal that the neuron will transmit downstream via its axon. This coincides with calculating the neuron's **activation** (or **post-activation**).

A 5-input example:



Pre-activation: $p = \sum_{i=1}^{5} w_{ij} x_{ij}$

Activation: $h = \Phi(p)$.

We will see different examples of activation functions Φ further on in the chapter.

The role of activation functions

The **activation function** Φ is a critical design element in neural networks since it introduces an element of **non-linearity**—without them, NNs are entirely useless (they are not stronger than a linear model).

We will better formalize this in a future chapter once we'll have introduced the matrix notation we'll use throughout these lectures.

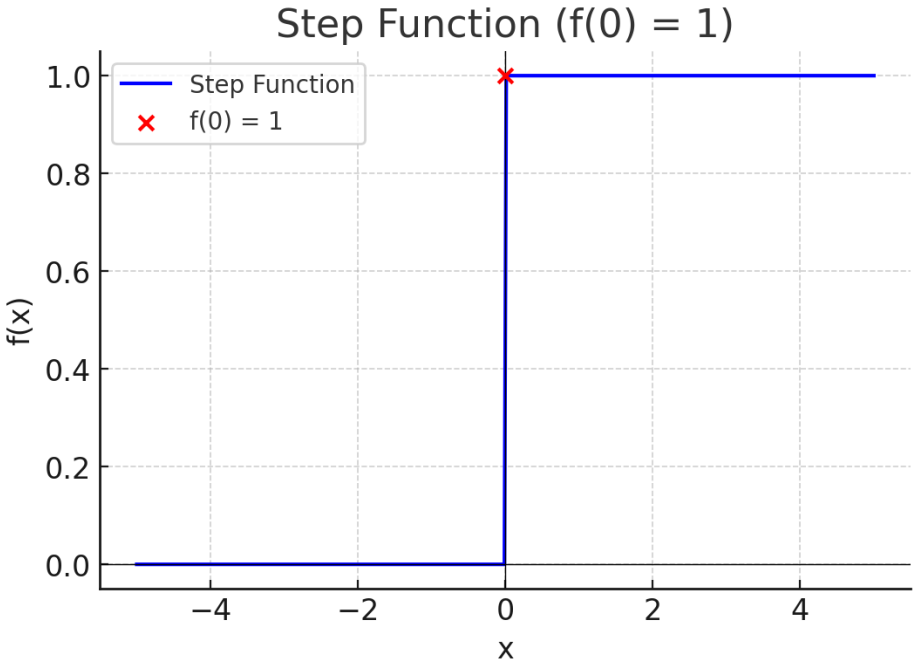
Let us note that, while the perceptron uses the *step* function, many other options exist, such as sigmoid and hyperbolic tangent functions.

The simplest activation function

The most simple activation function, the **step function**, is defined as follows:

$$\text{Step}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

Such a function is also known as the **Heaviside function**.

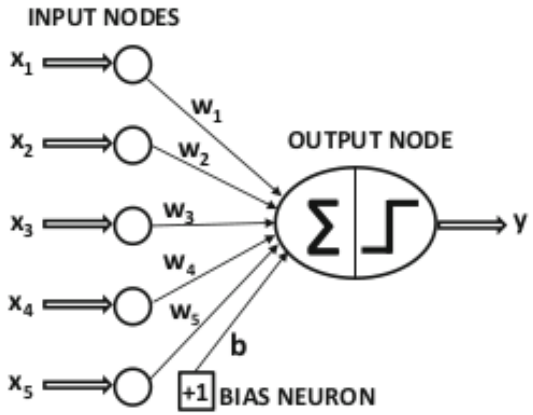


Adding a bias

So far, our neuron fires if $\sum_{i=1}^{n} w_{ij} x_i \geq 0$. We can make the firing threshold an arbirary value by introducing a **bias** parameter b.

For consistency with the notation we will use in the future, we use $-b$ as firing threshold, so that our neuron fires if $\sum_{i=1}^{n} w_{ij} x_i \geq -b$.

We can incorporate b in the McCulloch–Pitts neuron rather easily by adding an extra (dummy) input which is constant and always equal to 1, and to which we assign the synaptic weight b.



With this, the neuron's pre-activation becomes $p = \sum_{i=1}^{n} w_{ij} x_i + b$. In vector notation, we have $p = w^\top x + b$.

3. A Logical Calculus

Logical Computations

Logic was particularly influential in the intellectual climate of the 1940s.

In the background stood Whitehead and Russell's monumental *Principia Mathematica* (1910–1913), an ambitious project whose goal was to reconstruct all of mathematics within a formal logical system, treating mathematical reasoning as a special case of logical inference.

Closely related was the Hilbert program, which placed **formal logic** at the center of the foundations of mathematics. Its aim was to secure mathematics by expressing it in precise symbolic languages and proving, using logical reasoning, that these formal systems were **consistent** and **complete**. The program thus made logic not only a descriptive language for mathematics, but the primary tool for its justification.

Although Gödel's incompleteness theorems (1931) had already shown that sufficiently expressive formal systems cannot be both complete and provably consistent, the logic-centered conception of reasoning remained dominant. As a result, cognition itself was often understood as a form of logical computation—this is clear in McCulloch and Pitts's work.

In their 1943 work (*"A Logical Calculus of the Ideas Immanent in Nervous Activity"*), McCulloch and Pitts

- show how simple neural structures can implement all elementary Boolean-logic connectives (AND, OR, NOT);
- demonstrate that a neural network endowed with **cyclic connections** and sufficient memory can compute any computable function (which, in practice, means that a neural network with cyclic connections can execute any algorithm).

It should be noted that modern neural networks contain very little logic.

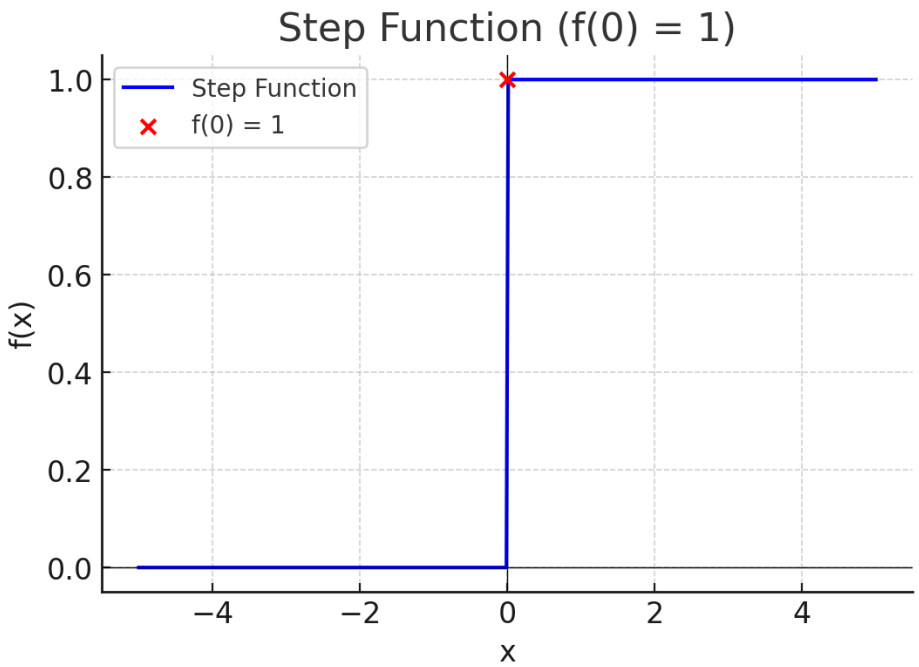
As we said, McCulloch and Pitts' focus on logic reflects the historical period in which their work was carried out, when logic was considered to be the **fundamental language of cognition**, even in psychology.

In their words, by demonstrating that their networks could implement **propositional logic**, McCulloch and Pitts provided (in their view) a *neurophysiological basis for reasoning*.

Let us try to build a single-neuron NN which realizes each of the operators NOT, AND, and OR.

We will use:

- $\Phi(x) = \text{Step}(x)$ (with $\text{Step}(0) = 1$)
- $x \in \{0, 1\}$



The NOT operator

x	$y = \mathrm{NOT}\, x$
1	0
0	1

Since $y = \mathsf{Step}(wx + b) = \begin{cases} 1 & \text{if } wx + b \geq 0 \\ 0 & \text{if } wx + b < 0 \end{cases}$, we need to find values for w and b such that

$$(x, y) = (1, 0) \Rightarrow \mathsf{Step}(w + b) = 0 \Leftrightarrow w + b < 0$$
$$(x, y) = (0, 1) \Rightarrow \mathsf{Step}(b) = 1 \Leftrightarrow b \geq 0.$$

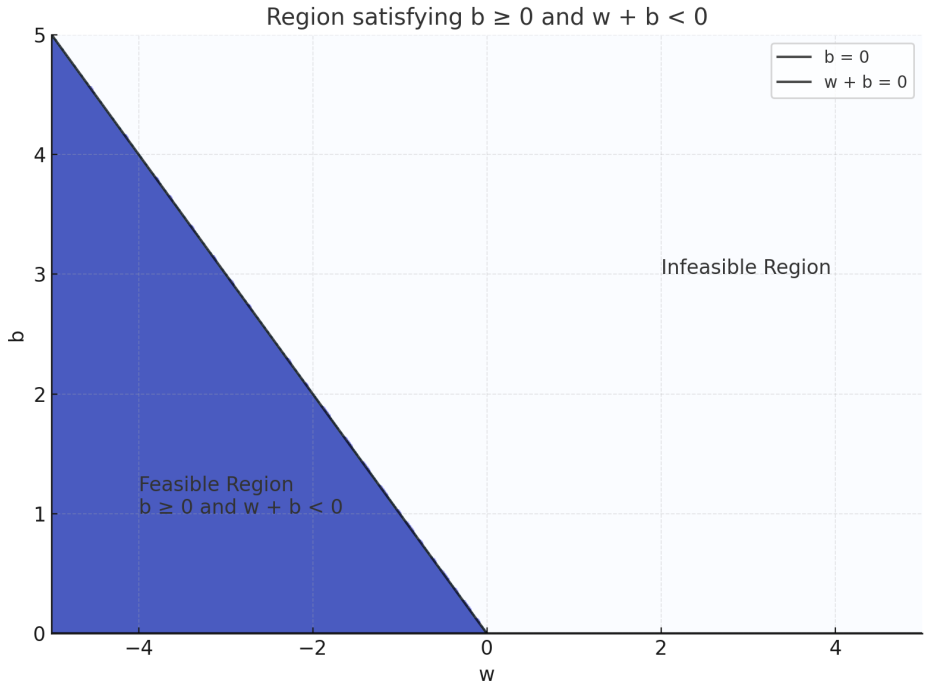
We obtain the two-inequality system

$$w + b < 0$$
$$b \geq 0.$$

The system

$$w + b < 0$$
$$b \geq 0.$$

has ∞^2-many solutions: infinitely-many with 2 degrees of freedom – one for each choice of w, b in the figure:



A feasible solution is $b = 1$, $w = -2$, which yields:

$$y = \text{Step}(-2x + 1).$$

We found

$$y = \mathsf{Step}(-2x + 1).$$

Let's test it:

$$\begin{aligned} x = 0 \Rightarrow \mathsf{Step}(1) \qquad\quad &= 1 \quad \checkmark \\ x = 1 \Rightarrow \mathsf{Step}(-2 + 1) \quad &= 0 \quad \checkmark. \end{aligned}$$

The AND operator

x_1	x_2	$y = x_1 \text{ AND } x_2$
0	0	0
0	1	0
1	0	0
1	1	1

Since $y = \text{Step}(w_1x_1 + w_2x_2 + b) = \begin{cases} 1 & \text{if } w_1x_1 + w_2x_2 + b \geq 0 \\ 0 & \text{if } w_1x_1 + w_2x_2 + b < 0 \end{cases}$, we need to find values for w and b such that

$$\begin{aligned}
(x_1, x_2, y) = (0,0,0) &\Rightarrow \text{Step}(b) = 0 \Leftrightarrow b < 0 \\
(x_1, x_2, y) = (0,1,0) &\Rightarrow \text{Step}(w_2 + b) = 0 \Leftrightarrow w_2 + b < 0 \\
(x_1, x_2, y) = (1,0,0) &\Rightarrow \text{Step}(w_1 + b) = 0 \Leftrightarrow w_1 + b < 0 \\
(x_1, x_2, y) = (1,1,1) &\Rightarrow \text{Step}(w_1 + w_2 + b) = 1 \Leftrightarrow w_1 + w_2 + b \geq 0.
\end{aligned}$$

We obtain the four-inequality system

$$\begin{aligned}
&b < 0 \\
&w_2 + b < 0 \\
&w_1 + b < 0 \\
&w_1 + w_2 + b \geq 0.
\end{aligned}$$

To solve

$$\begin{aligned} &b < 0 \\ &w_2 + b < 0 \\ &w_1 + b < 0 \\ &w_1 + w_2 + b \geq 0, \end{aligned}$$

let's (tentatively) pick $b = -2$. We obtain the system

$$\begin{cases} w_2 < 2 \\ w_1 < 2 \\ w_1 + w_2 \geq 2. \end{cases}$$

A feasible solution is $w_1 = w_2 = 1$, which yields:

$$y = \mathrm{Step}(x_1 + x_2 - 2).$$

Let's test it:

$$\begin{array}{lll} (x_1, x_2) = (0,0) \Rightarrow \mathrm{Step}(-2) & = 0 & \checkmark \\ (x_1, x_2) = (0,1) \Rightarrow \mathrm{Step}(1-2) & = 0 & \checkmark \\ (x_1, x_2) = (1,0) \Rightarrow \mathrm{Step}(1-2) & = 0 & \checkmark \\ (x_1, x_2) = (1,1) \Rightarrow \mathrm{Step}(1+1-2) & = 1 & \checkmark. \end{array}$$

The OR operator

x_1	x_2	$y = x_1 \,\mathrm{OR}\, x_2$
0	0	0
0	1	1
1	0	1
1	1	1

Since $y = \mathrm{Step}(w_1x_1 + w_2x_2 + b) = \begin{cases} 1 & \text{if } w_1x_1 + w_2x_2 + b \geq 0 \\ 0 & \text{if } w_1x_1 + w_2x_2 + b < 0 \end{cases}$, we need to find values for w and b such that

$$\begin{aligned}
&(x_1, x_2, y) = (0,0,0) \Rightarrow \mathrm{Step}(b) = 0 \Leftrightarrow b < 0 \\
&(x_1, x_2, y) = (0,1,1) \Rightarrow \mathrm{Step}(w_2 + b) = 0 \Leftrightarrow w_2 + b \geq 0 \\
&(x_1, x_2, y) = (1,0,1) \Rightarrow \mathrm{Step}(w_1 + b) = 0 \Leftrightarrow w_1 + b \geq 0 \\
&(x_1, x_2, y) = (1,1,1) \Rightarrow \mathrm{Step}(w_1 + w_2 + b) = 1 \Leftrightarrow w_1 + w_2 + b \geq 0.
\end{aligned}$$

We obtain the system

$$\begin{aligned}
&b < 0 \\
&w_2 + b \geq 0 \\
&w_1 + b \geq 0 \\
&w_1 + w_2 + b \geq 0.
\end{aligned}$$

Let's (tentatively) pick $b = -1$. We obtain the system

$$\begin{aligned} w_2 &\geq 1 \\ w_1 &\geq 1 \\ w_1 + w_2 &\geq 1. \end{aligned}$$

A feasible solution is $w_1 = w_2 = 1$, which yields:

$$y = \mathsf{Step}(x_1 + x_2 - 1).$$

Let's test it:

$$\begin{array}{lll} (x_1, x_2) = (0,0) \Rightarrow \mathsf{Step}(-1) & = 0 & \checkmark \\ (x_1, x_2) = (0,1) \Rightarrow \mathsf{Step}(1-1) & = 1 & \checkmark \\ (x_1, x_2) = (1,0) \Rightarrow \mathsf{Step}(1-1) & = 1 & \checkmark \\ (x_1, x_2) = (1,1) \Rightarrow \mathsf{Step}(1+1-1) & = 1 & \checkmark. \end{array}$$

A classical result

So far, we have shown that single-neuron NNs suffice to realize the three fundamental operators of Boolean algebra. Indeed:

Boolean representability theorem (McCulloch-Pitts (1943)): Given sufficiently-many neurons, a NN with a threshold activation function can represent any Boolean function.

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Learning

Let us remark that, in the McCulloch–Pitts model, the synaptic weights are fixed (set *a priori* by the designer).

This in contrast with modern NNs, where the synaptic weights are subject to learning.

In spite of this, the McCulloch and Pitt's model laid the groundwork for a later insight: that the **synaptic weights of a neural network should be subject to learning**.